Jieyu Zheng

Email: jzzheng@caltech.edu Website: http://jieyusz.github.io

EDUCATION

California Institute of Technology, Pasadena, U.S.A. Sep. 2020 - Present

Doctor of Philosophy in Neurobiology, Expected in Jun. 2026

Thesis topic: Complex Cognition in Mouse Maze Navigation, With and Without Cortex
Advisor: **Dr. Markus Meister**, Biaggini Professor of Biological Sciences
2024 Chen Innovator Grant Awardee
2023 Chen Diversity and Inclusion Grant Awardee

University of Cambridge, Cambridge, U.K. Oct. 2018 - Jul. 2019

Master of Philosophy in Psychology and Education (First Class). Advisor: Wendy Browne
Thesis Topic: Understanding Shame in Mathematical Achievement – A Systematic Review Using Meta-analysis

Cornell University, Ithaca, NY, U.S.A. Aug. 2016 - May 2018

Bachelor of Science in Biological Engineering, Magna Cum Laude (GPA:3.80/4.3)
College of Agriculture and Life Sciences (CALS) Dean's List (GPA above 3.50 Every Semester)
2018 Rhodes Scholarship in China Finalist

Shanghai Jiao Tong University (SJTU), Shanghai, China Sep. 2014 - Jun. 2016

Bachelor of Engineering in Food Science and Engineering | Zhiyuan Honor Degree and Scholarship (Top 5%)
GPA (overall): 3.91/4.3; ranking before transfer to Cornell: 1/162
China National Scholarship (Top 1%)

RESEARCH PROJECTS

Cognition With and Without Cortex: Mice in the Manhattan Maze [info] Dec. 2021 - Present

Advisors: **Markus Meister,** Professor of Biological Sciences; **Pietro Perona,** Professor of Electrical Engineering, Caltech

- Designed behavioral apparatus "the Manhattan Maze", experiments and built the arena for testing and recording.
- Managed, processed, and analyzed video data using computer vision tools and self-developed python packages.
- Applied to an independent neuroethology project for abstract reasoning (2024 Chen Innovator Grant).
- Led the maze group team (inc. 2 PhD student and 5 undergraduate research assistants) across two research groups.
- Advised the application of an online human virtual maze game study in the team.
- **Talks:** Harvard RL and Brain Seminar Fall 2024; Cognitive Computational Neuroscience 2024 (with Travel Award and selected talk, <5% of the abstracts).
- **Poster presentations:** Society for Neuroscience 2022; Curiosity, Creativity and Complexity 2023 (*with Travel Award*); Simons Collaboration on the Global Brain (SCGB 2023 site visit); HHMI Janelia Meeting 2025; Cognitive Computational Neuroscience 2025; Society for Neuroscience 2025 (*With TPDA award*).

The Unbearable Slowness of Being: Human behaviors at 10 bits/s [info] Mar. 2021 – Dec. 2025

Advisor: **Markus Meister,** Professor of Biological Sciences, Caltech

- Performed literature review and wrote the review of human behavioral studies as the first author
- **Talks:** Chen Institute Workshop on Cross-Species Modalities in Cognition and Behavior; Explore Caltech

Mesolimbic Dopamine Signaling and Cognitive Flexibility | *Research Assistant* Sep. 2019 - Feb. 2020

Advisor: Trevor Robbins, Professor of Cognitive Neuroscience, University of Cambridge

***Ex vivo* Imaging of *Drosophila* Olfactory System Development** | *Research Assistant* May - Aug. 2017

Advisor: Liqun Luo, Professor of Biology, Investigator of Howard Hughes Medical Institute, Stanford University

High Fat Diet and Alzheimer's Disease-related Pathology | *Research Assistant* Oct. 2016 - May 2018

Advisor: Chris Schaffer, Associate Professor of Meinig School of Biomedical Engineering, Cornell University

Functions of CXCL12 during Recovery from Ischemic Strokes in Mice | *Research Assistant* Jan. - Oct. 2015

Advisor: Yongting Wang, Professor of Med-X Neuroscience and Engineering Centre, SJTU

PUBLICATIONS [Google Scholar]

Zheng, J., and Meister, M. (2024). The unbearable slowness of being: Why do we live at 10 bits/s? *Neuron 11* (2), 192-204

Zheng, J., Turan, Z., Zeyu, J., Guimaraes, R., … Perona, P. and Meister, M. (In prep). Cognition with and without cortex: Rapid Learning, Long-term Memory and Generalization in the Manhattan Maze.

Zheng, J., Guimaraes, R., Hu J.Y., Perona P., and Meister, M. (2024). Mice in the Manhattan Maze: Rapid Learning, Flexible Routing and Generalization, With and Without Cortex. *Cognitive Computational Neuroscience, 2024*.

Guimaraes, R., … **Zheng, J.,** and Perona P. (2025). Humans and Mice Navigate Mazes Alike. Can AI Beat Them? *Cognitive Computational Neuroscience, 2025*.

Jiang, L., Li, W., Mamtilahun, M., Song, Y., Ma, Y., Qu, M., Lu, Y., He, X., **Zheng, J.** . . . Wang, Y. (2017). Optogenetic Inhibition of Striatal GABAergic Neuronal Activity Improves Outcomes After Ischemic Brain Injury. *Stroke, 48*(12), 3375-3383.

TEACHING AND ADVISING EXPERIENCES

Bi 23 Undergraduate Tutorial: The Ethology of Learning | *Instructor, Caltech* Spring 2025

- Independently developed the 10-week course content
- Supervised undergraduate field research projects and final presentations
- Hosted guest speakers for interdisciplinary discussions

CNS 187 Neural Computation | *Head Teaching Assistant* Spring 2022, 2023

Instructors: Markus Meister & Ueli Rutishauser, Professors of Computation & Neural Systems, Caltech

- Designed weekly homework assignments and final projects.
- Held weekly office hours and monitored online discussion forums.
- Oversaw course logistics, lecture recording and attendance.

President for the Neurotechers, Caltech 2023 - 2025

Academic Event Co-chair for the Neurotechers, Caltech Feb. 2022 - Jun. 2023

Data Science and AI for Neuroscience Summer School, Caltech | *Participant* Jul. 2022

Executive Education Programs at Møller Centre, University of Cambridge | *Client Relationship Assistant* Jul. - Sep. 2019

BEE 2600 Principles of Biological Engineering | *Undergraduate Teaching Assistant* Jan. - Dec. 2017

Cornell Cooperative Extension for Students with Special Needs | *Mentor* Feb. - May 2018

Harvard College AUSCR Summit for Young Leaders in China | *Exceptional Teaching Fellow* Aug. 2018

BEE 4890 Social Entrepreneurship with the SOS Children's Village in Chile | *Project Manager* Aug. - Dec. 2017

Cornell Empathy, Assistance and Referral Service (EARS) | *Peer Counsellor* Aug. - Dec. 2017